ST5227 - Applied Statistical Learning
Fall 2026

Data Science Project

Group Project

This will be a group assignment. Groups must have a minimum of 3 individuals and can have at most 4 individuals. You may find your own group members, and you can create your group on Canvas: People → Groups → Course Project. You will have one week to select your groups (deadline will be posted on Canvas). Those who are not in a group after the deadline will be randomly assigned into groups. After the deadline some groups may be merged or people may be added to your group.

After groups are set, you will have another week (again, deadline on Canvas) to select your dataset and fill in its information on the shared Excel sheet provided on Canvas. The dataset must be selected to satisfy what is detailed below. No two groups can analyze the same dataset. Search the excel sheet for an entry of the same dataset before entering yours. Do not alter other groups' entries (I will check change logs if I need to).

Assignment

You will practice the data science skillsets you have learned throughout the course in the analysis of a real dataset.

You will choose a real (and complicated) dataset to analyze using the tools and concepts you have learned in this course. You should treat the project like you would a job: Assume a client or your boss handed you this dataset and tasked you to perform an open-ended exploratory analysis. The dataset should match the scope of the course: it should be big and complicated; in particular, the dataset should have multiple variables and there should be a natural supervised learning (regression and/ or classification) task that one would be interested in solving. You should apply the statistical models and numerical experiments we have practiced in the labs to draw conclusions about the dataset, the problem, and the likely underlying mechanisms generating the data. Examples of key questions that should be explored (this list is not exhaustive):

1. What model(s) give you the best predictive performance on the supervised learning task?

2. Is there evidence to suggest the underlying data generating mechanism is nonlinear and/ or multivariate?

3. If model interpretability is very important, which model would you recommend, and what is the "price" you would pay (in terms of predictive performance) for

using that model instead of a better performing alternative (if any)?

4. Is there any dimensionality reduction and/ or clustering you could do on the data?

Your report will need to be concise. There will be a maximum page limit to the main report, and you will have to make careful choices on what to include and what to exclude in that report in order to meet the strict and short page limit and still make all the key points describing your findings and insights. You will also need to submit executable code. A short report is designed to be a challenge: It is surprisingly difficult to deliver a concise yet impactful statement, and it is hard not to showcase all the work you had done to get there. This reflects actual data science practice: When you deliver your findings, no one will ever read your 20 page detailed report – your busy boss will only ever read a short executive summary. Yet, you need to have the evidence ready to support your claims should they ever be challenged.

Instructions

1. See Canvas for due date and time.
2. Dataset.
   (a) You must select a dataset and fill it into the shared excel spreadsheet by the announced deadline. (See Canvas for both.) You must provide a URL where the dataset may be accessed. No two students can select the same dataset. First-come-first-served. You are required to search the excel sheet and determine if a previous entry has selected the same dataset.
   (b) There may be many datasets that go by the same name or look at the same problem. It is OK for two students to work on two such datasets, they just cannot be the same dataset or derived from the same underlying dataset. If in doubt, select something else.
   (c) You must conduct at least one supervised learning (i.e., regression and/ or classification) task, and so your selected dataset should accommodate this requirement.
   (d) The dataset must have at least $n = 200$ observations/ samples and at least $p = 5$ variables. If you find a dataset you are very interested in and wish to use that does not meet these requirements, you may request a special permission to still use the dataset.
   (e) There is no upper limit to the number of observations or variables, but datasets that are too large may be difficult to analyze for this project because it takes too long to conduct repeated experimental runs (e.g., as is common with cross-validation experiments). In this case you may downsample your dataset (or analyze only a subset), but the subset must be chosen in

an intelligent and data-driven way (e.g., by analyzing principal components, as in week 11, or feature importances, as in week 10). You must detail this downsampling/ subsetting procedure.

3. You must submit a main report that is no more than 8 pages in length at 12 point font, including figures and any references. Your main report should make all your key points and state all your findings with as much evidence as possible to back up your claims/ support your inferences. But you may additionally submit an unlimited number of supplementary pages to further support your findings, however, your supplementary material may not necessarily be considered in the marking of your work. **You will NOT be penalized for submitting a report much shorter than the page limit.**

4. Submit on Canvas. Your report may be in Word or PDF format. You must submit Python scripts (.py) or a jupyter notebook that can successfully execute all code and access to the dataset must be provided. I should be able to download that dataset and run your code with minimal changes (e.g., I can figure out how to change the code to point to the correct dataset location).

5. Note on the use of AI tools: NUS does not discourage the use of AI tools, but your report and code should be your work, not the work of an AI tool. **Remember:**

   - If the AI tool does the work implementing a skill you have never learned ⇒ you are **never-skilled**;
   - If the AI tool does the work and then you check that work after ⇒ you are **de-skilling**;
   - If *you do the work first* and then AI checks and critiques your work after ⇒ you are **up-skilling**.

   Obviously, up-skilling is always to be preferred. That's how we increase our expertise. But life is busy and full of deadlines, so we will often have to make hard choices. It is inevitable that we will de-skill on certain skills. Moreover, being never-skilled or de-skilled is not necessarily bad for all skills... I can only recommend that you always assess: "What skills do I want to learn?"

Figures and References

It would be very difficult to write an excellent report without figures. You likely will never produce such a report without figures (or at least such a report would not be good). Yet your figures should be carefully chosen to make impact but also to be concise. A picture is worth a thousand words. However, do not assume your figures will entirely speak for themselves. It must be completely clear what the figure is showing and the conclusions to be drawn from the figure. DO NOT assume that the reader is an expert on the analyses you are conducting.

You may apply any techniques from the course without references. Anything beyond the scope of this course should be supported by references, however, note that such analyses beyond the scope of this course are neither expected nor will be rewarded with additional marks. A report fully within the scope of this course and containing no references is entirely acceptable.

Data Sources

You may acquire your dataset from anywhere, but make sure you have the rights to use it for this assignment. I should also be able to access and download the dataset when assessing your work. There are many places to find good datasets, including the following:

- Kaggle: https://www.kaggle.com/datasets/
- UCI machine learning repository: https://archive.ics.uci.edu
- Google dataset search: https://datasetsearch.research.google.com

Project Requirements

There are 50 points total. These points will be re-weighted so that the project will account for 20% or your overall mark for the course. I will grade this as if I was your boss or client on the project.

1. Data Description (5 points)
   (a) Description of dataset, including number and nature (continuous, discrete, categorical, etc.) of the covariates and target(s). A discussion of whether missing values need to be dealt with. If only a subset of the dataset is analyzed, a description of the methodology for choosing the subset should be given and a description of the resulting subset should be given. Any methodology for selecting a subset of the covariates (if you do) should be considered with care so as not to discard the most important variables.
   (b) Modeling a supervised learning task is required, as such you should describe the task (and why someone would want to perform that task) and identify the appropriate covariates and targets.
2. Comparative numerical experiments (30 points) A numerical experiment comparing different regression/ classification methods should be conducted. These marks are broken down as follows:
   (a) A good selection of different models with different properties should be implemented so as to well explore different potential mechanisms/ structures underlying the dataset. (5 points)

(b) Comparisons should be robust; for example, multiple train/ test splits should be used and statistical significance (or lack thereof) in the differences between model performances should be explored. Methods should be clearly yet succinctly described. (5 points)

(c) Where necessary, you should ensure important hyper-parameters that significantly impact model performance are carefully selected, for example, with cross-validation. If a parameter is not expected to significantly impact performance, it is okay not to select it in a data driven way. When in doubt, justify your choices with cogent explanations. (5 points)

(d) The numerical results of your experiment should be analyzed and key conclusions drawn from those results should be described. These comprise the key findings of your analyses and form the primary takeaways. If your reader walks away remembering only one thing, it should be this. (15 points)

3. Feature exploration (5 points) Key features should be identified an explored. You may use any (and as many) feature selection/ investigation methods that you have learned. You may also restrict your attention to only a few covariates for close study, if you wish, in which case your methodology for that selection should be sound and clearly explained.

4. Unsupervised learning analysis (5 points) At least one supervised learning analysis should be conducted and findings from that analysis should be integrated into your report. (If you used an unsupervised learning methods to select a subset of a larger dataset to analyze for this project, that application will indeed satisfy this requirement.)

5. Limitations and potential criticisms (5 points) All analyses have limitations or weaknesses. You should discuss these. For example, if many of your key findings rely on a particular model, you should discuss the limitations of that model.

Grading Rubric

For each of the criteria/ requirements above, the 5 points will be awarded as follows:

- 5 points: An excellent treatment that is comprehensive and technically accurate. If applicable, experiments are tastefully chosen and code is clear and executes successfully. Key findings are clearly and succinctly communicated to demonstrate the key concepts emphasized in this course. Outputs are not obviously written by an AI model with no input, revision, or critical analysis by the student.

- 4 points: A very good treatment that demonstrates good command of the concepts in statistical modeling and analysis taught throughout this course. Most statements are technically accurate and methods are tastefully chosen, though one or two highly relevant techniques may have been excluded without clear reason.

Example: feature importance weights are presented and the key finding is communicated (i.e., the most important predictors are identified), but you do not provide a sense of robustness of those estimates (e.g., error bars over the weights), which are not entirely necessarily to support the key findings but would have demonstrated excellent data science practice.

- 3 points: A good treatment that just meets the expectations established throughout this course. Methods are varied and for the most part appropriately applied and their results correctly interpreted. There may be a few errors or misalignments, but a good command of the techniques and ability to perform statistical analyses is demonstrated. Code largely executes successfully and supports the claims made in the report, but there may be various parts of code that are unclear or missing. **Example:** You apply and analyze a variety of models, and your implementations and analyses are correct and impactful. However, you missed the opportunity to apply a GAM with visualizations of the impact of the top few most important features, which would have established the frontier of performance at the intersection of prediction and interpretability for the problem.

- 2 points: A treatment that falls below the expectations established throughout this course. The most appropriate methods are missing or are not correctly applied or their results not correctly interpreted in many cases. Code mostly does not execute successfully or may execute but does not clearly produce the evidence needed to support the claims in the report. **Example:** You implement a variety of models including linear models, black-box predictors, and a GAM, but you did not compare the predictive performance of the models and thereby draw analytical conclusions on the underlying structure of the data (e.g., linear vs nonlinear) and whether a simpler more interpretable model can be relied upon because a black box model provides no significant improvement in performance (or vice versa).

- 0 - 1 point: A poor treatment that does not meet expectations established throughout the course. Methods are not appropriately applied and findings are not correctly interpreted. Code is not clear and does not successfully execute to accomplish what is claimed in the report. This mark or zero marks will be given if I assess that an AI model was used with minimal care or thought by the student to add their independent work to the project. **Example:** The report and code are typical outputs of an AI model – overly verbose without making clear analytical contributions and with nonsensical code that happens to produce correct looking outputs.